

WASP-52b: EXHALE Atmospheric-Escape Model vs. He 10830 and H α Observations

EXHALE modeling notes

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Abstract

We model the upper atmosphere and escaping wind of the hot Jupiter WASP-52b with **EXHALE** (the extended ATES code) and compare the synthetic He I 10830 and H α transit transmission spectra against published observations. The setup follows Yan et al. (2022): the stellar XUV/FUV spectral energy distribution is taken from ϵ Eri (MUSCLES) scaled to the planet’s orbit, the lower boundary is pressure-anchored at $1\,\mu\text{bar}$, and the full Roche (tidal) potential is used. The steady-state mass-loss rates reproduce Yan et al. (2022) to within $\sim 10\%$ at every XUV level. The He 10830 absorption is shown to be governed by the metastable He 2^3S population: with the physically motivated L_1 -truncated domain, a He-poor composition ($\text{H}/\text{He} \simeq 98/2$) and the best-fit XUV flux ($0.5\,F_0$), the model reproduces the observed He 10830 line depth ($\sim 4\%$ vs. $\sim 3.4\%$). An extended $10\,R_p$ domain over-predicts the absorption by $3\text{--}10\times$ because of an unphysical He 2^3S tail beyond L_1 . The H α prediction is not trusted because the Ly α radiative transfer that sets the $n = 2$ hydrogen population is not solved here.

1 Target and goal

WASP-52b is a low-density hot Jupiter on a 1.75 d orbit around an active K2V star; excess absorption has been detected in both He 10830 (Kirk et al. 2022) and H α (Chen et al. 2020). Our goal is to compute a *base* EXHALE model (no fitting) and overplot the synthetic He 10830 and H α transmission spectra on the observations. The adopted planet/star parameters (Table 1) match Yan et al. (2022), whose hydrodynamic code (Yang & Guo 2018) we use as the physics reference.

Table 1: Adopted WASP-52 / WASP-52b parameters (Yan et al. 2022).

Quantity	Value	Note
Stellar mass $M_\star$	$0.87\,M_\odot$	K2V
Stellar radius $R_\star$	$0.79\,R_\odot$	
Stellar T_{eff}	5000 K	for the Balmer continuum
Planet mass M_p	$0.46\,M_J$	
Planet radius R_p	$1.27\,R_J$	base radius
Equilibrium temperature T_0	1304 K	= base temperature
Orbital distance a	0.0272 au	
He/H number ratio	0.087 (solar) ... 0.005	scanned (§7)
Base pressure	$1\,\mu\text{bar}$	$\rightarrow n_0 = 5.56 \times 10^{12}\,\text{cm}^{-3}$

2 Stellar input spectrum

WASP-52’s XUV spectrum is unobserved, so following Yan et al. (2022) we use the ϵ Eri spectrum from the MUSCLES Treasury Survey (France et al. 2016), scaled from Earth to the planet’s orbit by $(d_{\epsilon \text{ Eri}}/a)^2$ with $d_{\epsilon \text{ Eri}} = 3.212$ pc. The scaling reproduces Yan’s fiducial values:

Band	This work	Yan et al. (2022)
F_{XUV} (1–912 Å)	$3.44 \times 10^4 \text{ erg cm}^{-2} \text{ s}^{-1}$	3.42×10^4
$\beta_m = F(1\text{--}100)/F(1\text{--}912)$	0.218	0.22
FUV/NUV (912–2600 Å)	$2.63 \times 10^5 \text{ erg cm}^{-2} \text{ s}^{-1}$	$\sim 2.6 \times 10^5$

The FUV/NUV band photoionizes the $\text{He } 2^3S$ metastable level and is therefore essential for the He 10830 prediction. EXHALE reads the two-column (wavelength, flux) SED file directly and integrates it for the band luminosities and the frequency-dependent photoionization/heating rates.

Flux normalization (important). Yan et al. divide the substellar F_{XUV} by 4 to account for redistribution over the sphere, defining $F_0 \equiv F_{\text{XUV}}/4 = 8542 \text{ erg cm}^{-2} \text{ s}^{-1}$, and explore $0.25\text{--}8 F_0$. In EXHALE the SED is the full substellar flux and the 2-D approximation “Rate/2 + Mdot/2” multiplies it by 0.5 inside the simulation. Hence our $0.25\times$, $0.5\times$, $1.0\times$ fiducial SED map to Yan’s $0.5 F_0$, $1 F_0$, $2 F_0$ respectively. Yan’s best fit to the observations is $0.5 F_0$, i.e. our $0.25\times$ SED.

3 Numerical setup

The lower boundary is pressure-anchored at $1 \mu\text{bar}$ (**Base BC: pressure 1.0**), giving $n_0 = P/(k_B T_0) = 5.56 \times 10^{12} \text{ cm}^{-3}$, identical to Yan’s base condition. The $\text{He } 2^3S$ (metastable) network is enabled (**Include He23S? True**); trace metals (C/N/O+Mg/Ca/Na/Fe) are optionally added via **metals.inp**. Convergence uses the two-stage PLM→WENO3 march with a du-keyed hand-off to the Newton/JFNK solver. The automatic initial-condition selector (**IC mode: auto**) chooses a transonic wind seed (the interior cold sonic point sits at $r_c \simeq 2.42 R_p$).

Domain: tidal potential, two outer boundaries. The full Roche potential along the substellar axis is used in all runs. We compare two outer boundaries:

- **Extended ($10 R_p$):** matching Yan’s domain extent. This required a one-line extension so that an explicit **Outer radius** is honored in Roche mode while keeping the tidal terms (documented in the user manual / update log).
- **L_1 -truncated ($\simeq 2.53 R_p$):** the default Roche behavior, truncating at the inner Lagrange point where the effective gravity vanishes.

The tidal force is treated identically to Yan/Yang: the momentum equation carries gravity + stellar tide + centrifugal terms. The escape-energy drop from R_p to L_1 in the full Roche form (0.431 in GM_p/R_p units) agrees with the Erkaev et al. (2007) reduced form $K(\lambda) = 1 - 3/2\lambda + 1/2\lambda^3 = 0.428$ to 0.6% for WASP-52b, so the two tidal prescriptions are effectively equivalent here.

4 Mass-loss validation

At matched XUV flux the EXHALE steady-state mass-loss rates reproduce Yan et al. (2022, their Table 4) to within $\sim 10\%$ (Table 2). This confirms that the wind hydrodynamics and energy balance are consistent with the reference code; differences in the line predictions below are therefore not driven by the mass loss.

Table 2: Steady-state mass-loss rate $\log_{10} \dot{M}$ [g s^{-1}] vs. Yan et al. (2022) Table 4 ($\beta_m = 0.22$). Our SED multiple maps to Yan’s F_0 multiple as noted.

Composition	Flux	EXHALE	Yan (2022)
H/He = 92/8	$1 F_0$ (our $0.5\times$)	11.69	11.70
H/He = 92/8	$2 F_0$ (our $1.0\times$)	11.93	11.96
H/He = 98/2	$1 F_0$ (our $0.5\times$)	11.74	11.69
H/He = 98/2	$2 F_0$ (our $1.0\times$)	11.98	11.95
H/He = 92/8	$0.5 F_0$ (our $0.25\times$)	11.47	11.44
H/He = 98/2	$0.5 F_0$ (our $0.25\times$)	11.49	11.41

5 He 10830: extended domain over-predicts

With the $10 R_p$ extended domain the model He 10830 excess absorption is 15–50%, far above the observed $\sim 3.4\%$ (Fig. 1, left). The $\text{H}\alpha$ panel is shown but is not trusted (see §8).

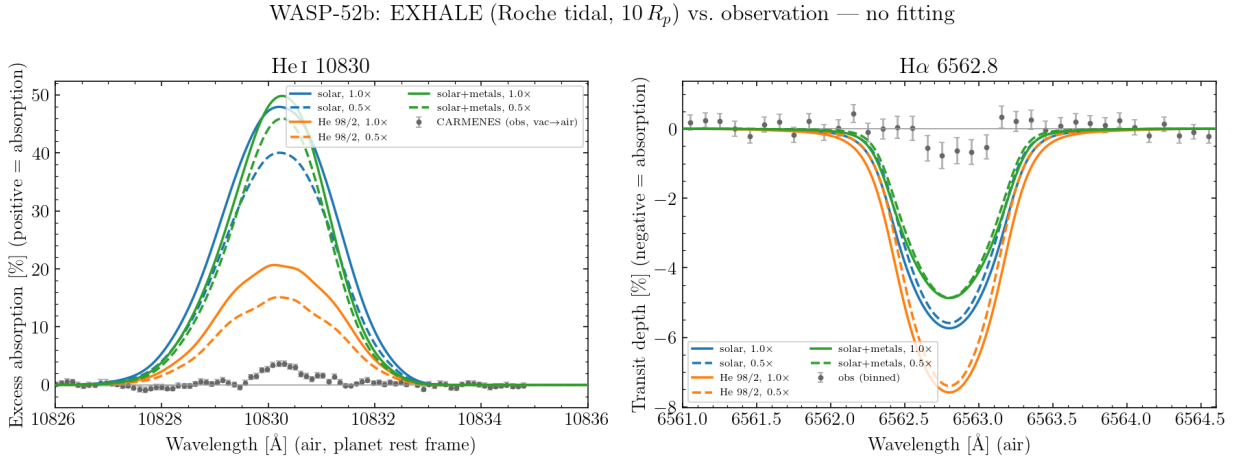


Figure 1: Extended-domain ($10 R_p$) model vs. observations. He 10830 (left, excess absorption, positive = absorption; observation converted vacuum→air) and $\text{H}\alpha$ (right, transit spectrum, negative dip = absorption). Color = composition, line style = XUV level. The model over-predicts He 10830 by 3–10 $\times$.

Diagnosis: the metastable He 2^3S tail. The over-prediction is *not* a mass-loss or flux-normalization error (the $\dot{M}$ matches Yan, §4). It arises from the spatial distribution of the He 2^3S population. In

the steady state

$$\frac{n(\text{He } 2^3S)}{n_{\text{He}^+}} \simeq \frac{\alpha_3 n_e}{\Phi_3 + A_{31} + (q_{31a} + q_{31b})n_e + Q_{31}n_{\text{H}}}, \quad (1)$$

where α_3 is the He^+ recombination into 2^3S , Φ_3 the FUV photoionization rate, $A_{31} = 1.27 \times 10^{-4} \text{ s}^{-1}$ the radiative decay, and q_{31}, Q_{31} the electron / hydrogen collisional sinks. We verify $\Phi_3 = 0.061 \text{ s}^{-1}$ (at $1 F_0$) directly from the SED and the NIST cross section, consistent with the FUV flux above. In the cool, rarefied outer wind the collisional sinks vanish, leaving photoionization as the only sink; the ratio $n(\text{He } 2^3S)/n_{\text{He}^+} = \alpha_3 n_e / \Phi_3$ then *rises* outward and the $\text{He } 2^3S$ density stays nearly flat from 2 to $5 R_p$. Because the transit depth is weighted by the impact parameter ($\propto r^2$), this sustained outer tail dominates: **72% of the He absorbing column comes from $r > L_1$ ($2.53 R_p$)**. Beyond L_1 the gas is Roche-unbound and physically funnels through L_1 rather than forming a complete spherical shell, so the 1-D spherical treatment to $10 R_p$ over-counts the absorption.

6 He 10830: L_1 -truncated domain reproduces the observation

Truncating the domain at L_1 removes the unphysical outer tail and brings the He 10830 depth into agreement with the data (Table 3, Fig. 2). At the best-fit flux ($0.5 F_0$) and a He-poor composition ($\text{H}/\text{He} = 98/2$), the model gives 3.9% vs. the observed $\sim 3.4\%$. The solar ($92/8$) composition remains too strong (7.4%), independently confirming Yan’s conclusion that a high H/He ratio is required to match the He 10830 line.

Table 3: He 10830 line-center excess absorption [%] for the extended ($10 R_p$) and L_1 -truncated domains. Observed $\sim 3.4\%$.

Flux	Composition	$10 R_p$	L_1 -trunc.
$0.5 F_0$	92/8 (solar)	30.6	7.4
$0.5 F_0$	98/2 (He-poor)	11.2	3.9
$1 F_0$	98/2	15.1	4.3
$2 F_0$	98/2	20.7	5.1

7 H/He scan

To locate the composition that matches the He 10830 line we scan the He/H number ratio from solar ($92/8$) to strongly He-poor ($99.5/0.5$) at the best-fit flux ($0.5 F_0$) in the L_1 -truncated domain (Table 4, Fig. 4). The He 10830 excess absorption decreases monotonically with decreasing He/H, from 7.4% (solar) to 1.5% ($99.5/0.5$). The observed depth ($\sim 3.4\%$) is bracketed by $\text{H}/\text{He} = 98/2$ (3.9%) and $99/1$ (2.5%), implying a best-fit ratio of $\text{H}/\text{He} \approx 98.3/1.7$ ($\text{He}/\text{H} \approx 0.017$). This is a strongly He-depleted atmosphere relative to solar, in line with the high H/He ratio that Yan et al. (2022) required to reproduce the He 10830 line.

8 $\text{H}\alpha$ caveat

The $\text{H}\alpha$ line absorbs from the $n = 2$ hydrogen population, which is set primarily by $\text{Ly}\alpha$ radiative pumping. EXHALE/TPM.py estimate the $\text{Ly}\alpha$ mean intensity with the simple Huang et al. (2017) prescription rather than a full $\text{Ly}\alpha$ radiative-transfer solution. The resulting $n = 2$ population, and

WASP-52b: EXHALE (Roche tidal, L_1 -truncated $\simeq 2.53 R_p$) vs. observation — no fitting

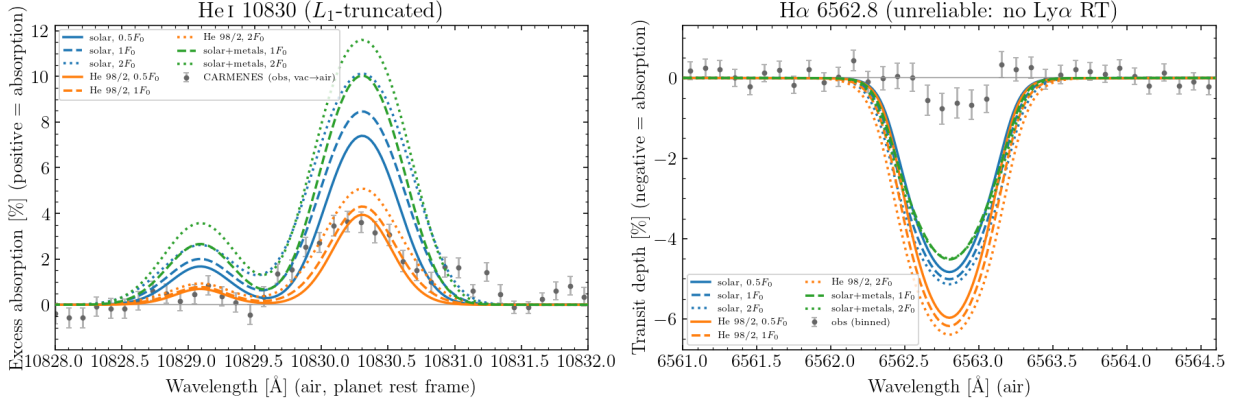


Figure 2: L_1 -truncated ($\simeq 2.53 R_p$) model vs. observations, same layout as Fig. 1. The He-poor $0.5 F_0$ model (orange solid) matches the observed He 10830 line. $H\alpha$ (right) is shown for completeness only.

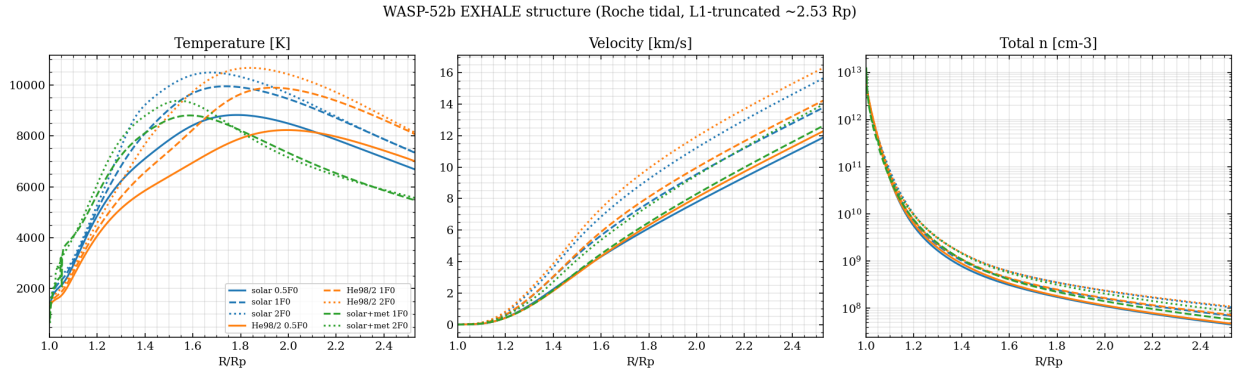


Figure 3: Converged atmospheric structure (temperature, velocity, total number density) for the L_1 -truncated runs. Color = composition, line style = XUV level.

hence the $H\alpha$ prediction, is therefore not reliable and is shown in the figures for completeness only. A proper $Ly\alpha$ RT treatment is required before the $H\alpha$ comparison can be interpreted.

9 Conclusions

1. EXHALE reproduces the Yan et al. (2022) steady-state mass-loss rates at every XUV flux to within $\sim 10\%$ (Table 2).
2. The He 10830 line depth is controlled by the metastable $He 2^3S$ population. An extended $10 R_p$ domain over-predicts the absorption by $3\text{--}10\times$ because the outer $He 2^3S$ tail beyond L_1 dominates the impact-parameter integral; the bound atmosphere ($r < L_1$) alone gives the observed depth.
3. With the L_1 -truncated domain, a He-poor composition ($H/He \simeq 98/2$) and the best-fit flux ($0.5 F_0$), the model reproduces the observed He 10830 line ($\sim 3.9\%$ vs. $\sim 3.4\%$), independently

Table 4: He 10830 line-center excess absorption [%] vs. composition, at $0.5 F_0$ in the L_1 -truncated domain. Observed $\sim 3.4\%$.

He/H	H/He	He 10830 [%]
0.087	92/8 (solar)	7.40
0.050	95.2/4.8	6.04
0.020	98/2	3.94
0.010	99/1	2.52
0.005	99.5/0.5	1.47

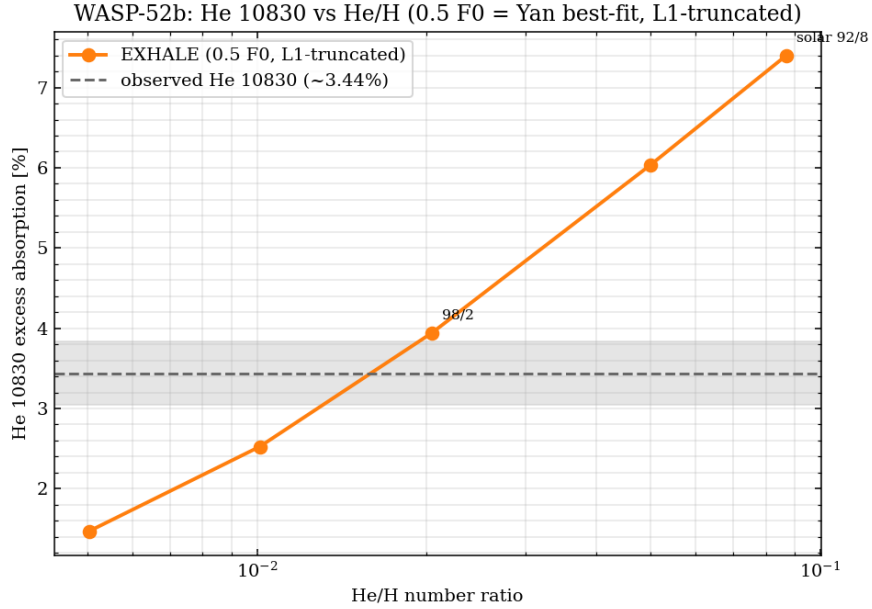


Figure 4: He 10830 excess absorption vs. He/H number ratio ($0.5 F_0$, L_1 -truncated). The dashed line and gray band are the observed depth ($3.44 \pm 0.4\%$). The model crosses the observation near $H/He \approx 98/2$, confirming the need for a He-depleted atmosphere.

confirming Yan’s requirement of a high H/He ratio.

4. $H\alpha$ is not modeled reliably here because the $Ly\alpha$ radiative transfer is not solved.

References

- Caldioli, A., Haardt, F., Gallo, E., et al. 2021, A&A, 655, A30. “Irradiation-driven escape of primordial planetary atmospheres. I. The ATES photoionization hydrodynamics code.”
- Erkaev, N. V., Kulikov, Yu. N., Lammer, H., et al. 2007, A&A, 472, 329. “Roche lobe effects on the atmospheric loss from hot Jupiters.”
- France, K., Loyd, R. O. P., Youngblood, A., et al. 2016, ApJ, 820, 89. “The MUSCLES Treasury Survey. I.”
- Huang, C., Arras, P., Christie, D., & Li, Z.-Y. 2017, ApJ, 851, 150. “A Model of the $H\alpha$ and Na

Transmission Spectrum of HD 189733b.”

Yan, D., Seon, K.-i., Guo, J., Chen, G., & Li, L. 2022, ApJ, 936, 177. “Modeling the H α and He 10830 Transmission Spectrum of WASP-52b.”

Yang, L., & Guo, J. 2018, RAA, 18, 077 (arXiv:1806.04719). “Effect of Orbital Distance on the Atmospheric Escape of Exoplanets.”